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$$\begin{aligned} & \int x \ln x dx \\ & \left\{ \begin{array}{l} u = \ln x \\ dv = x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \frac{1}{x} dx \\ v = \frac{x^2}{2} \end{array} \right. \\ & \int x \ln x dx \\ &= \frac{x^2 \ln x}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \\ &= \frac{x^2 \ln x}{2} - \int \frac{x}{2} dx \\ &= \frac{x^2 \ln x}{2} - \frac{1}{2} \cdot \frac{x^2}{2} + C \\ &= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C \end{aligned}$$

References